From Byte Tokens to Verified Tool Use: A Reproducible Local Language-Model Lab

Advanced Machine Learning Project
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Abstract

We connect a byte-level tokenizer, a decoder Transformer, pretraining, systems measurements, data cleaning, scaling extrapolation, post-training, and verified tool use in a locally reproducible experiment collection. A from-scratch 98.3-million-parameter model is trained for 1,500 updates on Apple M4, reducing fixed-window validation cross-entropy from 8.9119 to 2.7503 while still producing repetitive text. Fifteen smaller models support a fixed-token-budget scaling experiment: a largest-model hold-out prediction has 0.8575% relative error. On a separate constrained numeric task, same-base supervised fine-tuning, direct preference optimization, and verifier-reward policy gradients reach mean held-out accuracies of 90%, 90%, and 85%. A learned three-action policy completes new 16-goal ledger episodes averaging 68.08 tool actions. Self-review adds no success gain in ordinary episodes, but repairs deliberately corrupted draft actions at twice the inference-call cost. The experiments cover finite task domains and short training budgets on one Apple GPU. Code, measurements, data snapshots, small model bundles, screenshots, and training checkpoints accompany the report.

1 Problem and motivation

An implementation can run without establishing that its mathematics, learning signal, and evaluation agree. Our central problem is to build an inspectable chain from raw text to externally verified actions. We ask three questions: (1) can custom tokenizer and model components support 100-million-scale parameter updates; (2) can controlled local comparisons separate implementation speed, model-size effects, and training signals; and (3) can a learned policy complete dependent tool actions without using an answer-producing rule at inference time?

The public *from-token-to-agent* project (jackmcgradylee, 2026) informed the tokenizer and size-pilot comparisons. Stanford CS336 provides related systems, scaling, and alignment assignment specifications (Stanford CS336, 2026).

2 Implementation and experimental protocol

2.1 Tokenizer and decoder

The tokenizer implements Byte Pair Encoding (BPE), beginning with 256 byte symbols and an End of Sequence (EOS) symbol. Pair frequencies are counted within Unicode-aware pieces; deterministic tie-breaking and ranked, non-overlapping merges make saved rules reproducible. Encoding followed by decoding preserves input text, including Chinese and emoji.

The decoder contains explicit linear projections, Root Mean Square Normalization (RMSNorm), Rotary Position Embedding (RoPE), causal Multi-Head Attention (MHA), Swish Gated Linear Units (SwiGLU), residual connections, and tied input/output embeddings. Future positions are masked.

The token-level objective is

$$\mathcal{L}_{\mathrm{LM}} = -\frac{1}{T}\sum_{t=1}^{T} \log p_\theta(x_{t+1} \mid x_{\le t}). \tag{1}$$

PyTorch supplies tensors, automatic differentiation, and AdamW. Tests compare attention and cross-entropy to reference PyTorch operators, check causality and gradients, overfit a fixed batch, and compare uninterrupted CPU training with normal checkpoint restoration.

2.2 Large-model run

The architecture has vocabulary capacity 8,192, width 768, 12 layers, 12 attention heads, and feed-forward width 2,304: 98,323,200 parameters. The TinyStories snapshot contains 512 training stories and 128 validation stories, with documented whitespace normalization and CDLA-Sharing-1.0 attribution (Eldan and Li, 2023). Only training text fits the tokenizer; its realized vocabulary is 5,696. The streams contain 172,991 and 39,574 tokens.

Training uses Metal Performance Shaders (MPS), 32-bit floating point, batch size one, context 128, and 1,500 AdamW updates. The learning rate peaks at $3 \cdot 10^{-4}$ after 50 warmup updates and follows cosine decay. The sampled budget is 192,000 tokens. Evaluation uses 16 fixed, potentially overlapping 128-token windows, not the full validation set. A prior separate 100-update run demonstrates stop-and-resume behavior; it is not concatenated with the new run.

2.3 Systems measurements

We compare explicit scaled dot-product attention against PyTorch Scaled Dot-Product Attention (SDPA), including forward and backward passes. We also implement a per-layer Key-Value (KV) cache with offset-correct RoPE, chunked prefill, batch checks, and explicit context-overflow errors. Cached logits are checked against full-prefix logits on CPU; greedy outputs match on MPS.

After other training processes finish, measurements synchronize the GPU before and after wall-clock timing, exclude two warmups, and report medians and ranges. Attention uses seven repeats; decoding uses five. No Triton kernel or multi-GPU experiment is reported: the selected host has one Apple GPU and no compatible Triton/CUDA test device.

2.4 Raw-data and scaling experiment

We save a MediaWiki XML export of 16 specified Simple English Wikipedia articles, including revision URLs and a raw-file hash. Cleaning removes templates, references, markup, and selected tables; malformed template input is counted explicitly. Exact hashing and five-word-shingle Jaccard similarity at threshold 0.8 remove duplicates before an article-level training/validation split.

Fourteen articles remain: ten training and four validation articles. All size pilots share a 512-symbol training-only BPE tokenizer, context 64, two layers, four heads, batch eight, and 100 updates. Widths 32, 48, 64, 80, and 96 produce 43,168–289,248 parameters. Each size is independently trained with seeds 42, 43, and 44, using 51,200 sampled tokens. Fixed validation windows are shared across runs.

Following the empirical power-law approach of Kaplan et al. (2020), we fit

$$L(N) = a + b(N/N_0)^{-\alpha} \tag{2}$$

to the four smallest size means. A bounded grid over α and offset a, with least-squares amplitude b, is fit without the largest size. This is an iso-data, short-training size study, not a compute-optimal joint data/model law.

2.5 Same-base post-training

All alignment branches start directly from the same 289,248-parameter width-96 seed-42 checkpoint, not the 98M story model. The task is $\max(a, b)$ for digits 0–9. A fixed permutation splits the 100 ordered pairs into 80 training and 20 test pairs. The base and all methods use the same ten-digit output candidate mask. Each branch initializes directly from the common checkpoint.

Supervised Fine-Tuning (SFT) minimizes answer-only negative log likelihood. Direct Preference Optimization (DPO) uses the true answer and a uniformly sampled wrong digit with a frozen reference (Rafailov et al., 2023):

$$-\log\sigma\left(\beta\left[\log\frac{\pi_\theta(y^+\mid x)}{\pi_0(y^+\mid x)}-\log\frac{\pi_\theta(y^-\mid x)}{\pi_0(y^-\mid x)}\right]\right),\qquad \beta=0.5. \tag{3}$$

Reinforcement Learning with Verifiable Rewards (RLVR) samples eight one-token actions per prompt, with exact correctness reward r_i. A leave-one-out baseline gives $A_i = r_i - \frac{1}{7}\sum_{j\neq i} r_j$. REINFORCE (Williams, 1992) minimizes

$$-\frac{1}{8}\sum_i A_i \log\pi_\theta(y_i\mid x) + 0.02\, D_{\mathrm{KL}}(\pi_\theta\|\pi_0). \tag{4}$$

Each method uses 250 updates, batches of 32, and three seeds. Update/prompt budgets match; wall time and rollout costs do not.

2.6 Tool environment, training, and self-review

The resettable ledger environment presents one current integer and target at a time. Legal tools decrement (`L`), increment (`R`), and commit (`S`). Commit advances to the next goal only when the value equals its target; invalid commits, range violations, and exhausted budgets terminate with explicit failure. External verification checks all final values and ordered commits.

Starting from the stage-four SFT seed-42 model, agent training uses 10,556 demonstration/review rows from verified trajectories of 1, 2, 4, and 8 goals. The Transformer predicts one of three action tokens. It receives 300 imitation updates followed by 24 sampled environment episodes with terminal verified rewards and a fixed 0.5 baseline. The policy-gradient update sums action log probabilities, with entropy coefficient 0.001. The terminal rewards in these 24 episodes are all positive; imitation training already produces successful behavior.

Self-review asks the same trained model for an action after showing its draft. It does not call the demonstration rule or the success verifier to choose a correction. Evaluation uses 12 new-seed episodes at each of 1, 4, 8, and 16 goals, with a 128-action budget. Goal sequences are new; digit states are not a held-out domain. We separately corrupt 10% of draft actions to test review robustness, rather than mixing that setting into ordinary success rates.

3 Results

3.1 Tokenization, training, and throughput

On the same validation text, character tokenization uses 89,501 units, whitespace-preserving word tokenization uses 35,330 pieces with 982 unseen pieces, and byte BPE uses 39,574 units without an unknown-word placeholder. These are length/coverage measurements, not comparable language-model losses across different vocabularies.

The 98M model reduces fixed-window validation cross-entropy from 8.9119 to 2.7503. Summed optimizer-step time is 495.45 seconds; this excludes preparation, evaluation, and checkpoint I/O. Greedy text changes from frequent-token repetition to short learned phrases, but remains repetitive and ungrammatical.

Attention medians for lengths 64/128/256 are 1.0329/1.1378/1.1286 ms for explicit attention and 0.9403/1.1120/1.1071 ms for SDPA; observed forward/gradient errors are zero. Timing ranges overlap, so these modest ratios should not be generalized. With 32 prompt tokens and 64 new tokens, decoding falls from 431.405 to 390.203 ms (1.106×), with identical output and 97,280 cache bytes. The two-layer width-64 benchmark model is untrained and measures computation, not text quality.

3.2 Scaling and post-training

The fitted offset is 3.328899, amplitude 1.077007, exponent 0.33, and $N_0 = 43,168$. The held-out prediction is 3.903813 versus observed 3.870621: 0.8575% relative error.

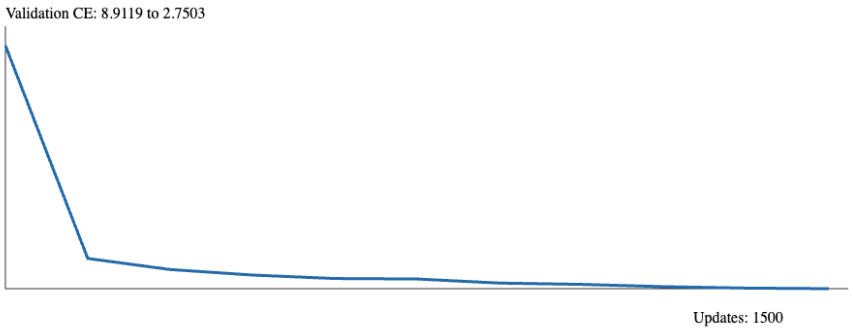


Figure 1: Fixed-window validation cross-entropy during the 98M/1,500-update run.

Table 1: Fixed-budget size means over three seeds. The last row is excluded from fitting.

Parameters	Validation CE	Use
43,168	4.4095	Fit
84,720	4.1777	Fit
139,584	4.0768	Fit
207,760	3.9630	Fit
289,248	3.8706	Hold-out

The base scores 5%. Each RLVR seed samples 64,000 answers. The split places the unique answer-zero pair $(0, 0)$ in the test set; zero is therefore absent from positive training labels. All branches miss that pair. Other errors include SFT predicting 5 for $\max(6, 5)$ and RLVR predicting 2 for $\max(2, 8)$. The 20-question test set limits the precision of method comparisons.

3.3 Verified agent and self-review

Successful 16-goal episodes average 68.08 tool actions. Ordinary self-review does not improve success and doubles calls from 68.08 to 136.17 on average. Its robustness benefit is measured under the specified draft-action corruption process. A saved eight-goal self-review trajectory completes 32 actions and matches every final target; its full observations, drafts, actions, and feedback are replayable.

4 Limitations and reproducibility

The 98M data budget is small and its generations remain weak. Fixed validation windows overlap and are not full-set evaluations. The XML collection contains only 16 selected articles, and two are discarded for malformed templates. Size pilots are far from convergence; a good local prediction does not validate a billion-scale extrapolation. Alignment uses single-token constrained responses on a tiny domain. Agent evaluation reuses known digit states in new goal sequences; it is not open-ended filesystem, web, or software-engineering agency. It shows a learned finite-domain controller executing long action chains. There is no measured incremental agent reward-learning benefit beyond imitation here, and no Triton or multi-GPU result.

Every stage includes commands, JSON records, and experiment screenshots. Data bundles include attribution, transformations, and SHA-256 manifests. The two stage-one checkpoints are approximately 1.18 GB each and include optimizer and random states. A replay demo reloads a neural policy and checks final environment states. The repository is available at `https://github.com/zongyangbigpolo/AdvancedMachineLearningTask`.

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Table 2: Constrained numeric-task accuracy. Times cover training loops only, averaged over seeds.

Method	Seed 42	Seed 43	Seed 44	Mean	Seconds
SFT	85%	95%	90%	90%	2.751
DPO	95%	90%	85%	90%	3.599
RLVR	85%	85%	85%	85%	3.418

Table 3: Successful episodes out of 12 per horizon. Noise denotes a separate 10% draft-action corruption test.

Policy / setting	1 goal	4 goals	8 goals	16 goals
Before agent training	0	0	0	0
Trajectory SFT	12	12	12	12
SFT + environment RL	12	12	12	12
SFT + RL + self-review	12	12	12	12
Noise, no review	9	2	2	0
Noise, self-review	12	12	12	12

References

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